In [5]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [ ]:

In [6]:
```
df=pd.read_csv('diabete.csv')
df.head()
```

Out[6]:

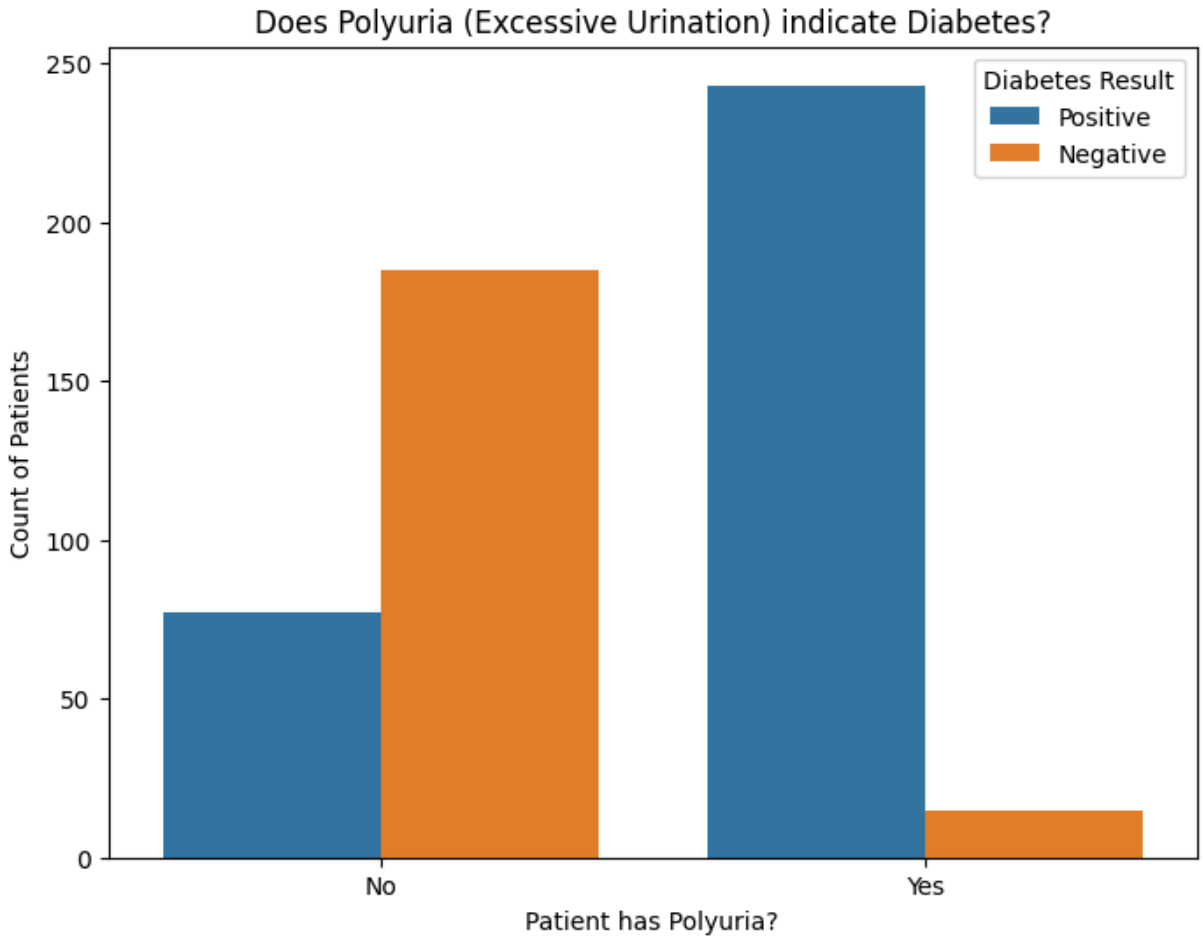
	Age	Gender	Polyuria	Polydipsia	sudden weight loss	weakness	Polyphagia	Genital thrush	visual blurring	It
0	40	Male	No	Yes	No	Yes	No	No	No	
1	58	Male	No	No	No	Yes	No	No	Yes	
2	41	Male	Yes	No	No	Yes	Yes	No	No	
3	45	Male	No	No	Yes	Yes	Yes	Yes	No	
4	60	Male	Yes	Yes	Yes	Yes	Yes	No	Yes	

In [7]:
```
mapping={
    'Yes':1,'No':0,
    'Positive':1,'Negative': 0,
    'Male': 1, 'Female': 0
}
```

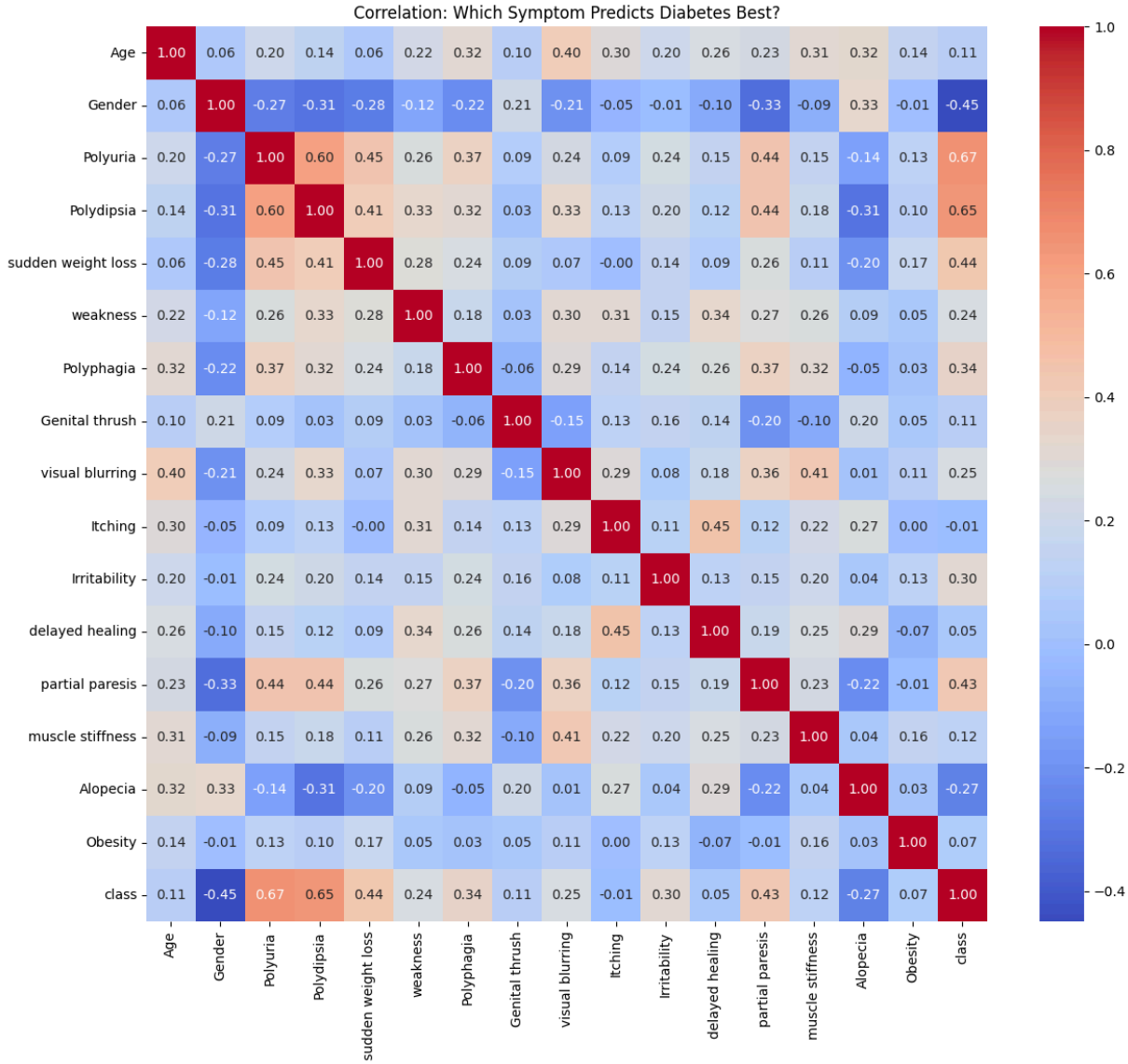
In [8]:
```
df_numeric=df.replace(mapping)
```

```
C:\Users\krish\AppData\Local\Temp\ipykernel_28020\75638357.py:1: FutureWarning: Down
casting behavior in `replace` is deprecated and will be removed in a future version.
To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To o
pt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', Tru
e)`
  df_numeric=df.replace(mapping)
```

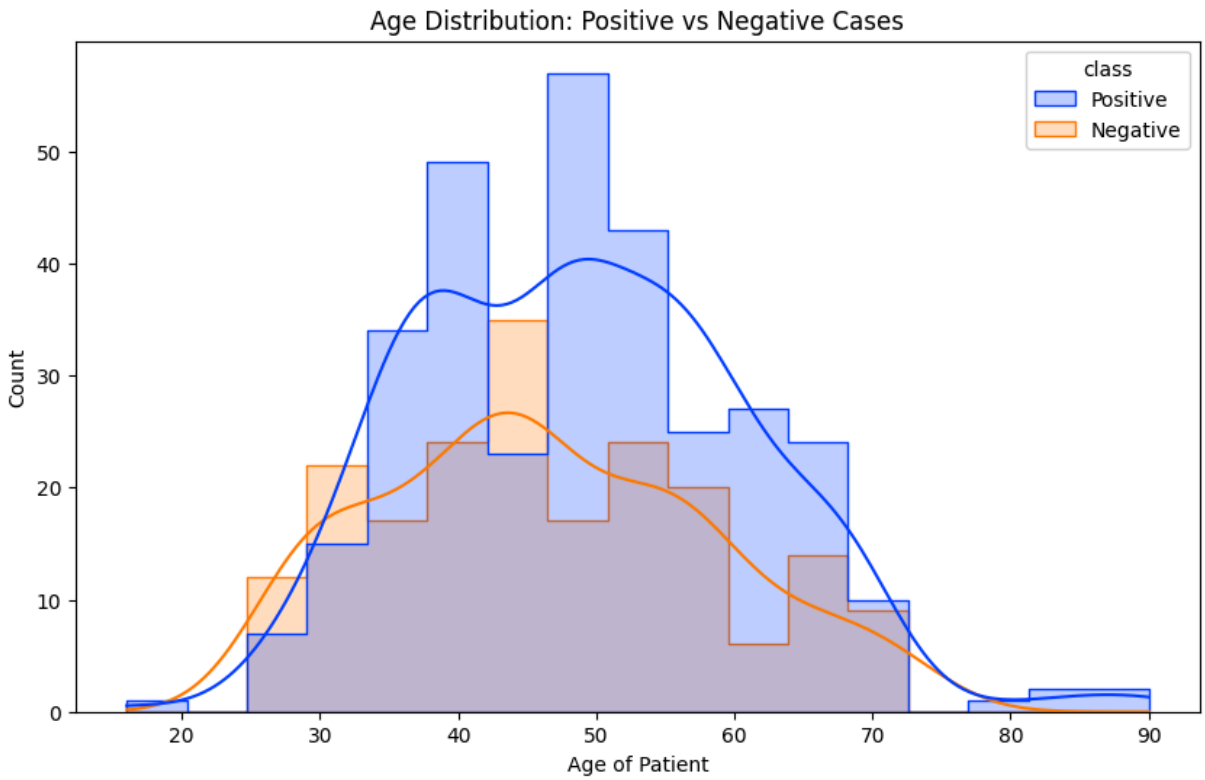
In [16]:
```
plt.figure(figsize=(8,6))
sns.countplot(data=df,x='Polyuria',hue='class')
plt.title('Does Polyuria (Excessive Urination) indicate Diabetes?')
plt.xlabel('Patient has Polyuria?')
plt.ylabel('Count of Patients')
plt.legend(title='Diabetes Result')
plt.show()
```



In [17]:
```
plt.figure(figsize=(14, 12))
sns.heatmap(df_numeric.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation: Which Symptom Predicts Diabetes Best?')
plt.show()
```

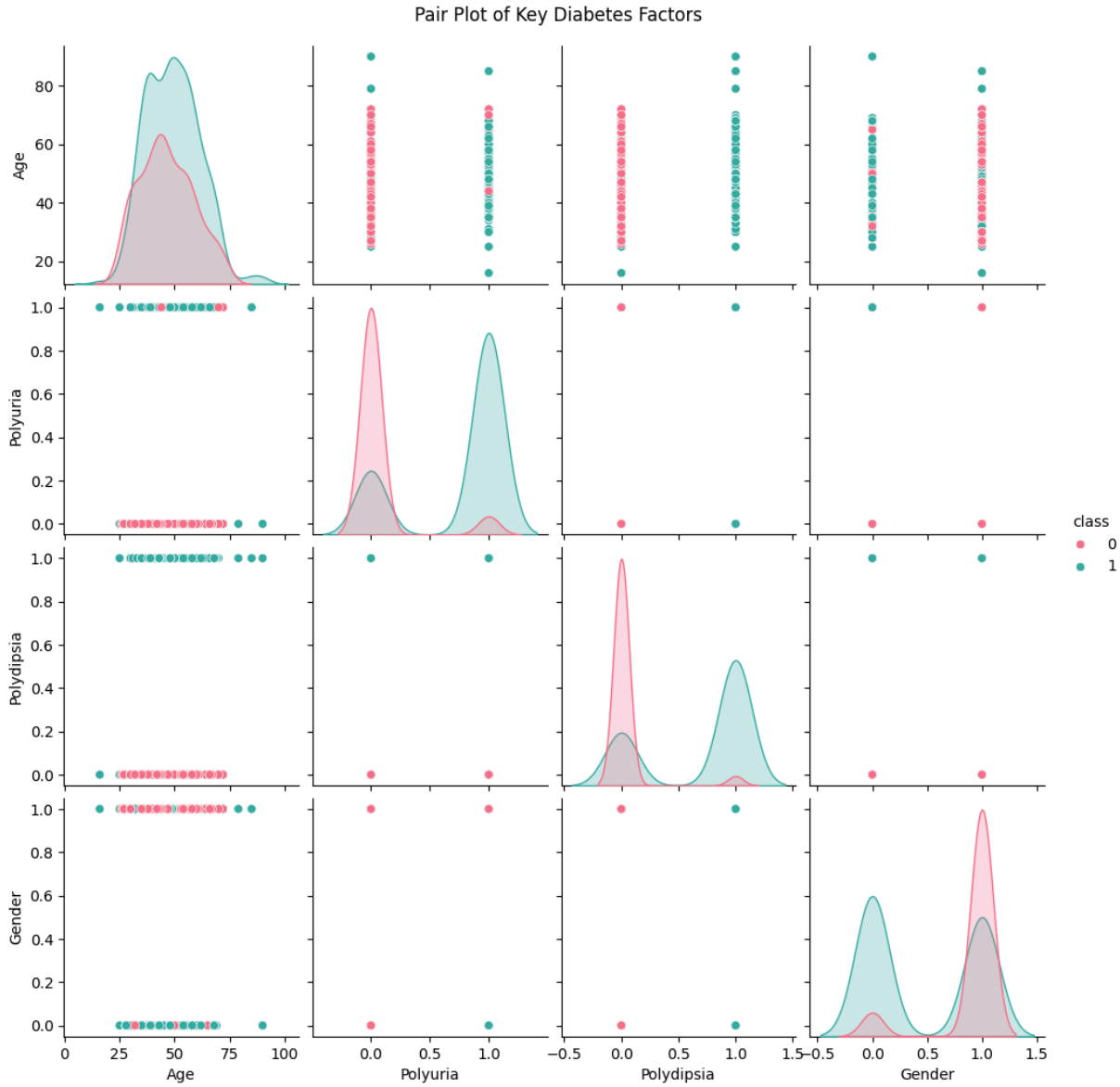


In [18]:
```
plt.figure(figsize=(10, 6))

# Histogram of Age, colored by Result
sns.histplot(data=df, x='Age', hue='class', kde=True, element="step", palette='brig

plt.title('Age Distribution: Positive vs Negative Cases')
plt.xlabel('Age of Patient')
plt.show()
```

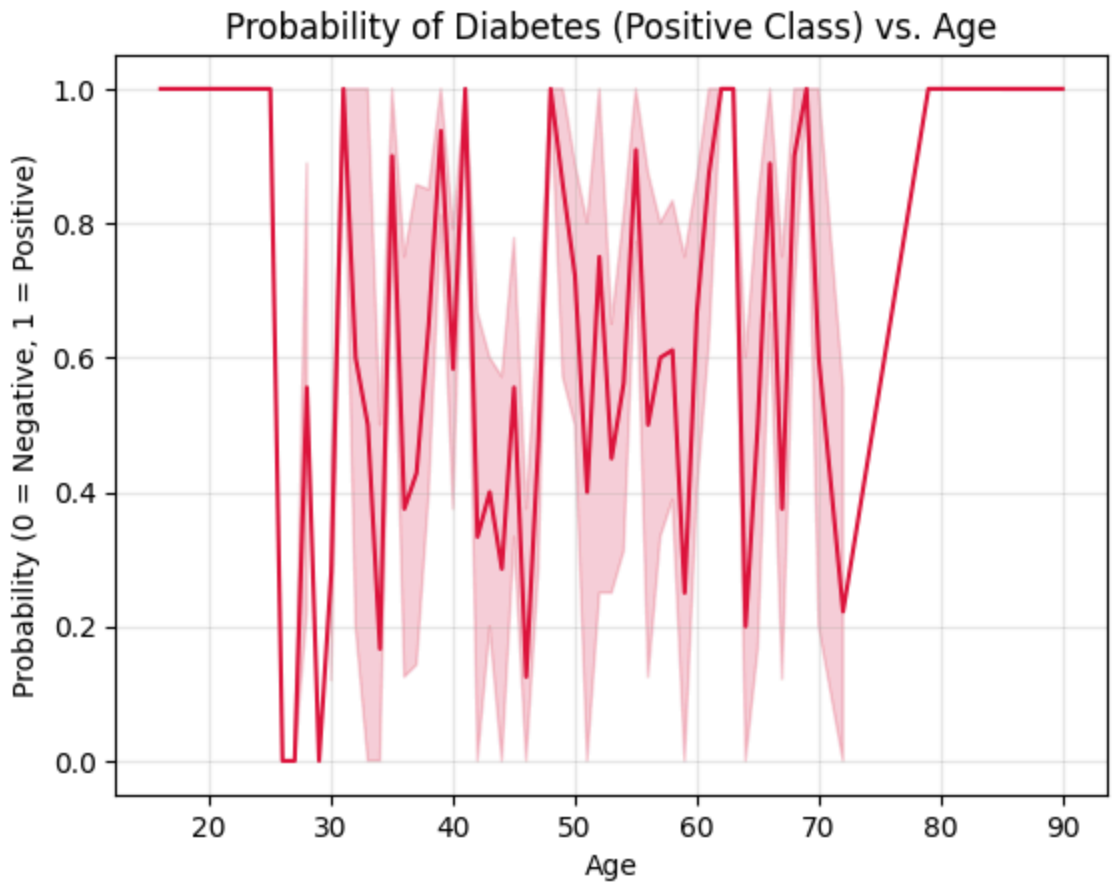
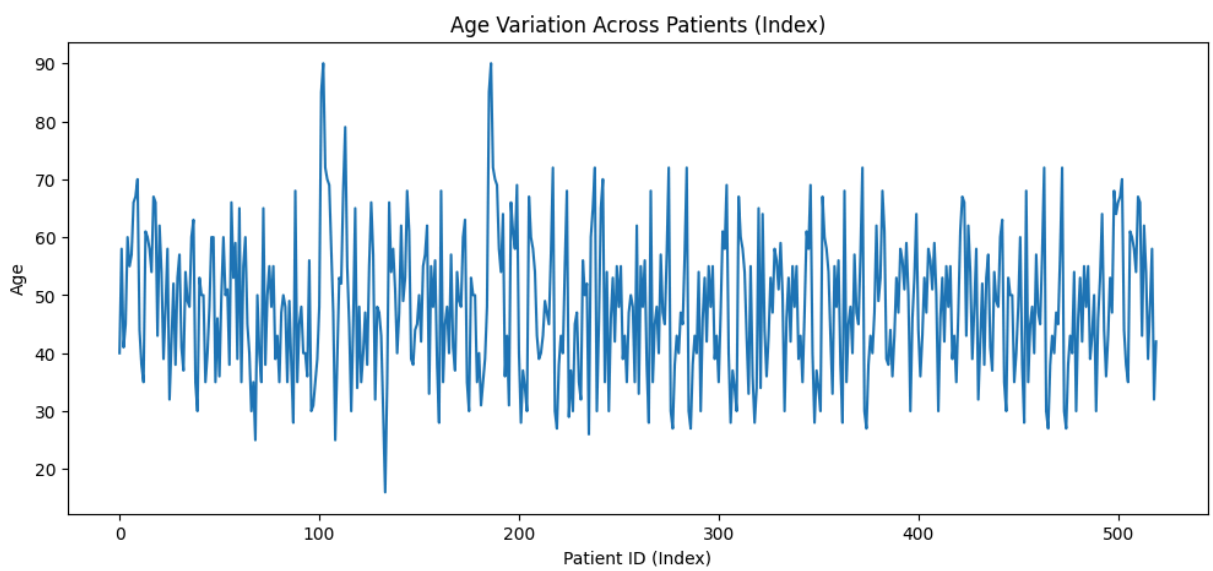


In [20]:
```
# Select only the most important features to keep the plot readable
# 'class' is included to color the dots (hue)
selected_cols = ['Age', 'Polyuria', 'Polydipsia', 'Gender', 'class']

plt.figure(figsize=(10, 8))

# We use the numeric dataframe we created earlier
sns.pairplot(df_numeric[selected_cols], hue='class', diag_kind='kde', palette='husl

plt.suptitle('Pair Plot of Key Diabetes Factors', y=1.02)
plt.show()
```

```
<Figure size 1000x800 with 0 Axes>
```



In [21]:
```
sns.lineplot(data=df_numeric, x='Age', y='class', color='crimson')

plt.title('Probability of Diabetes (Positive Class) vs. Age')
plt.ylabel('Probability (0 = Negative, 1 = Positive)')
plt.grid(True, alpha=0.3)
plt.show()
```



In [22]:
```
plt.figure(figsize=(12, 5))
sns.lineplot(data=df['Age'])
plt.title('Age Variation Across Patients (Index)')
plt.xlabel('Patient ID (Index)')
plt.ylabel('Age')
plt.show()
```